

AIRLINE PASSENGER SATISFACTION ANALYSIS

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INTRODUCTION

In today's fast-paced aviation landscape, the pursuit of passenger satisfaction stands as a cornerstone for airlines aiming to excel in service delivery and maintain a competitive edge.

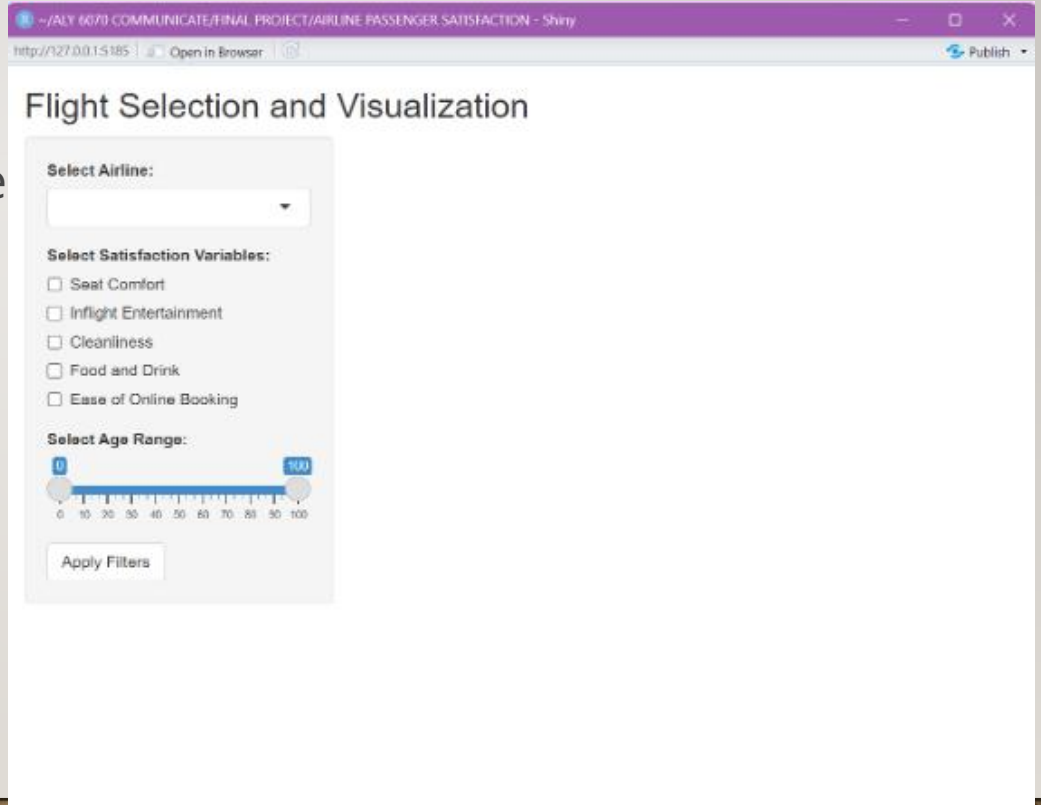
In this analysis, we leveraged various platforms such as Shiny R, Tableau, and Power BI.

The Airline Passenger Satisfaction dataset was sourced from Kaggle.



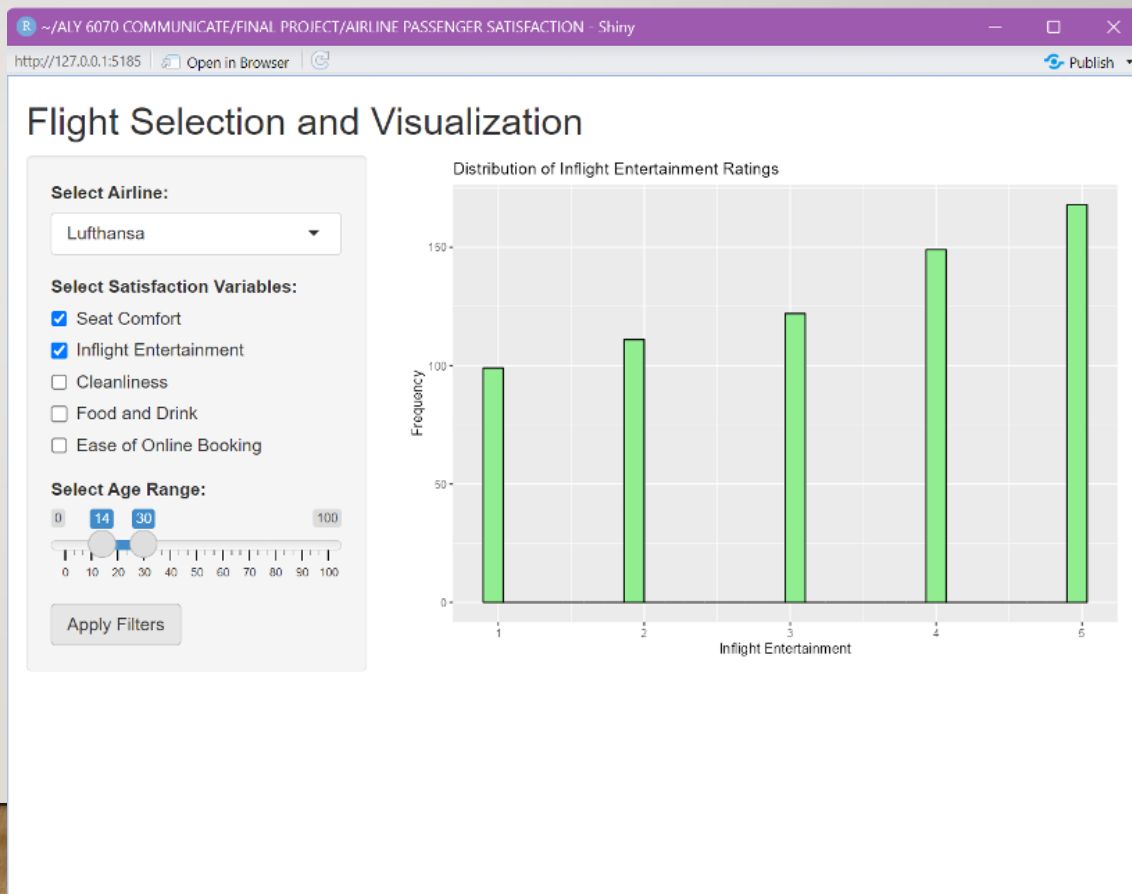
SHINY R

Tab showing comparison by certain satisfaction variables like age, airline, and other factors.



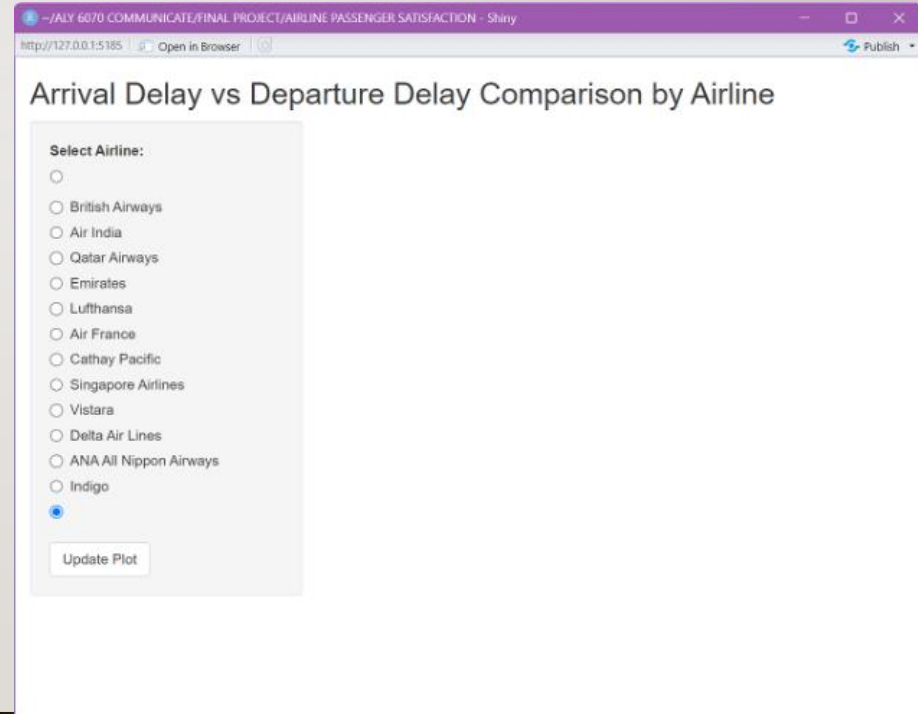
SHINY R

After selecting certain factors



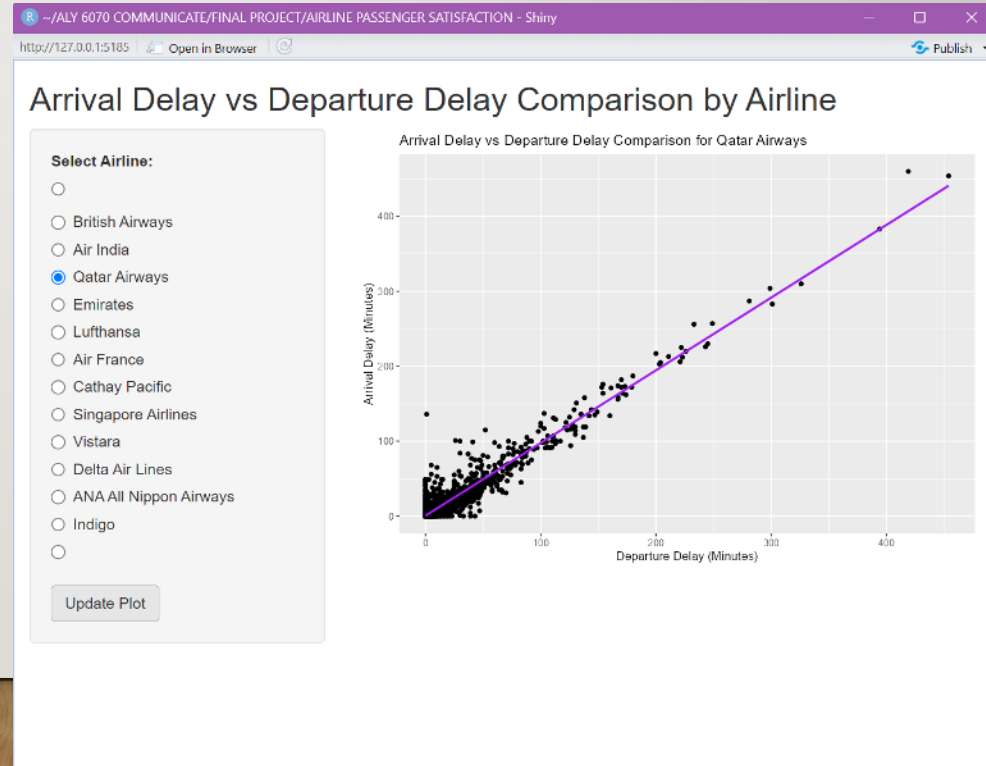
SHINY R

Tab showing comparison of arrival and departure delay by airline name.



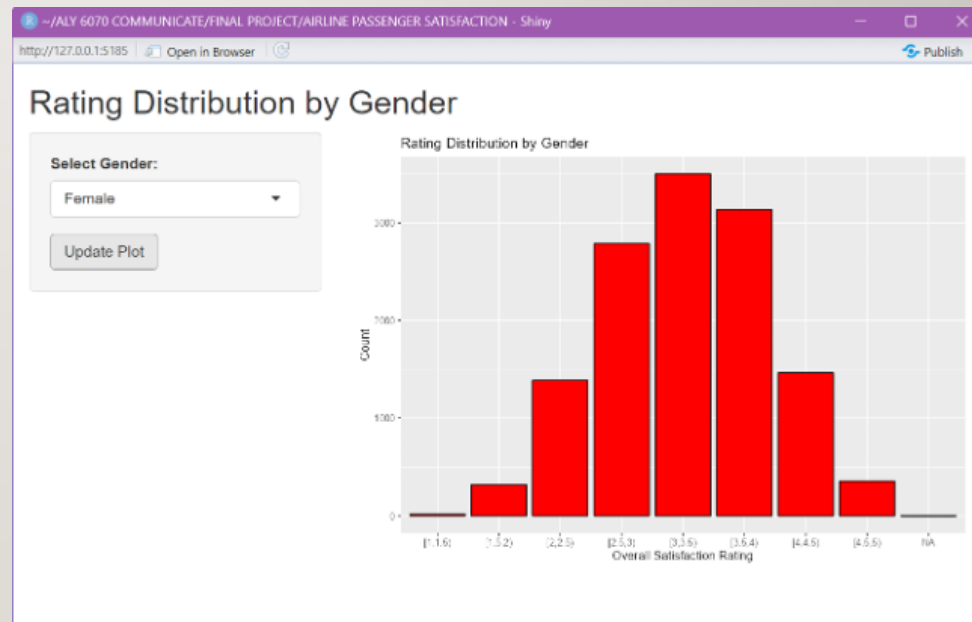
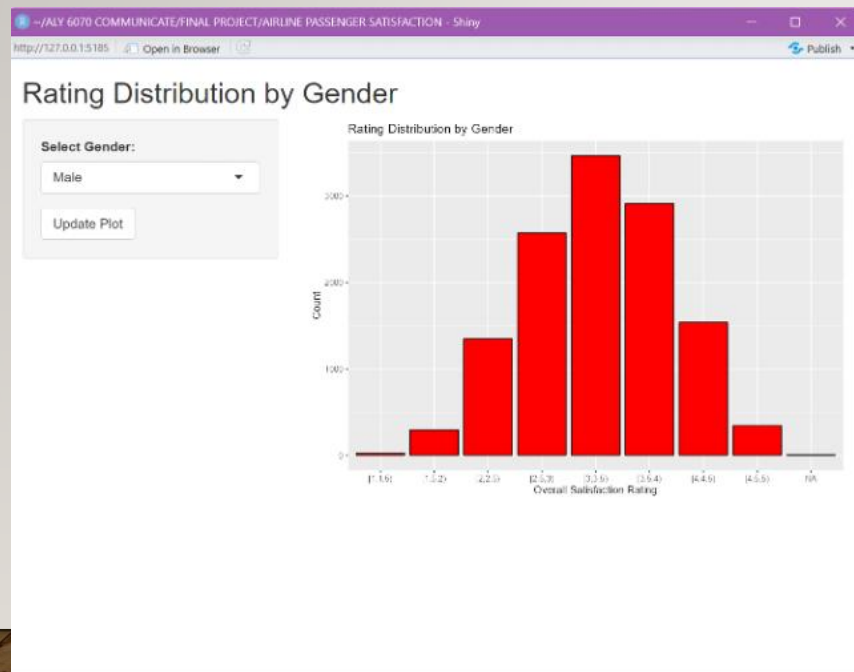
SHINY R

After selecting a certain Airlines



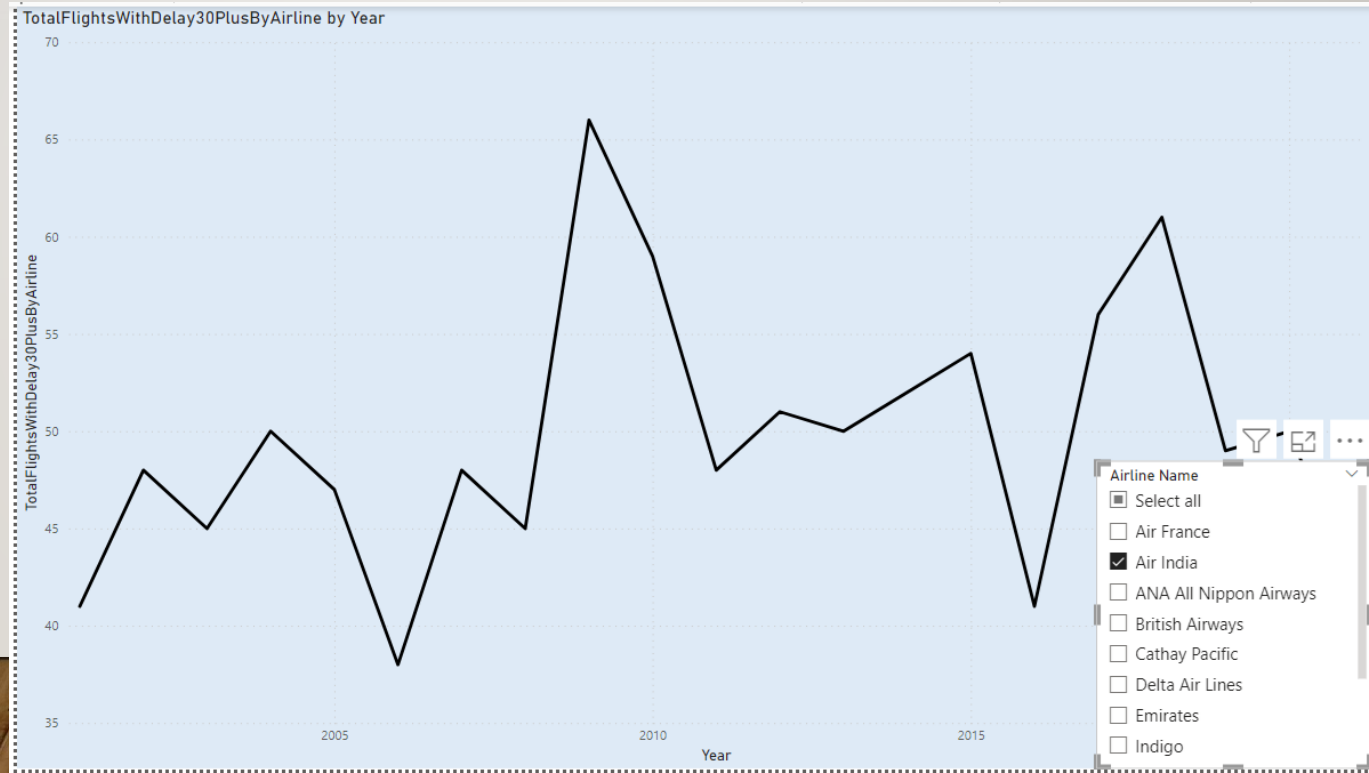
SHINY R

Rating Distribution by gender



POWERBI

Line chart showing total flights with delays greater than 30 minutes over the years.



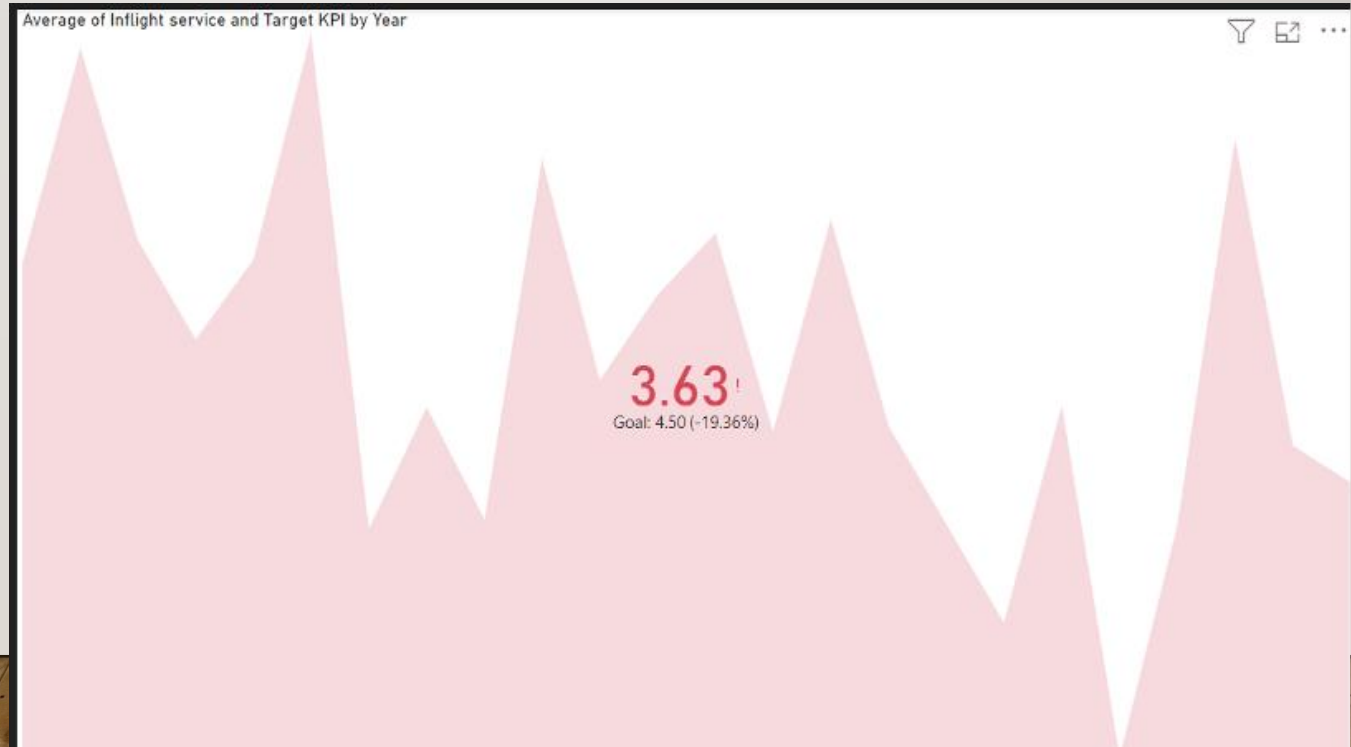
POWERBI

DAX formula which compares the total flights with delays of more than 30 minutes, filtered by airline name.

```
1 TotalFlightsWithDelay30PlusByAirline =  
2 SUMX(  
3     VALUES('Airline Passenger Satisfaction'[Airline Name]),  
4     CALCULATE(  
5         COUNTROWS('Airline Passenger Satisfaction'),  
6         'Airline Passenger Satisfaction'[Departure Delay in Minutes] > 30  
7     )  
8 )
```

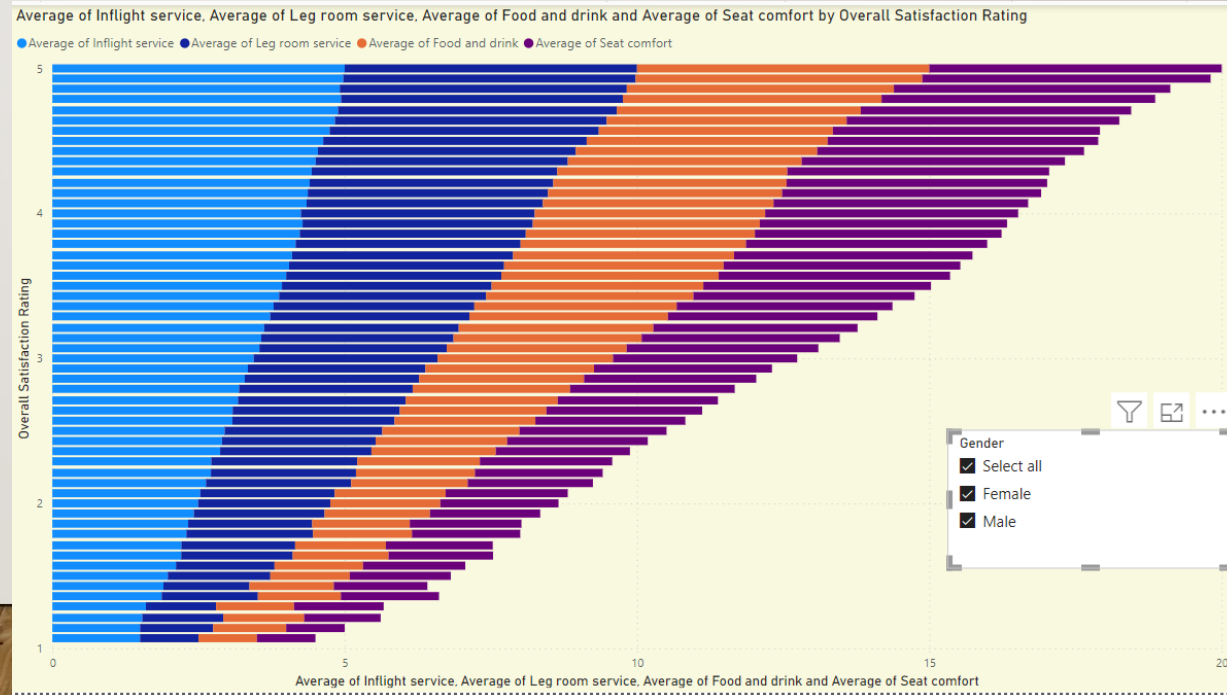
POWERBI

KPI of average of in-flight services by year



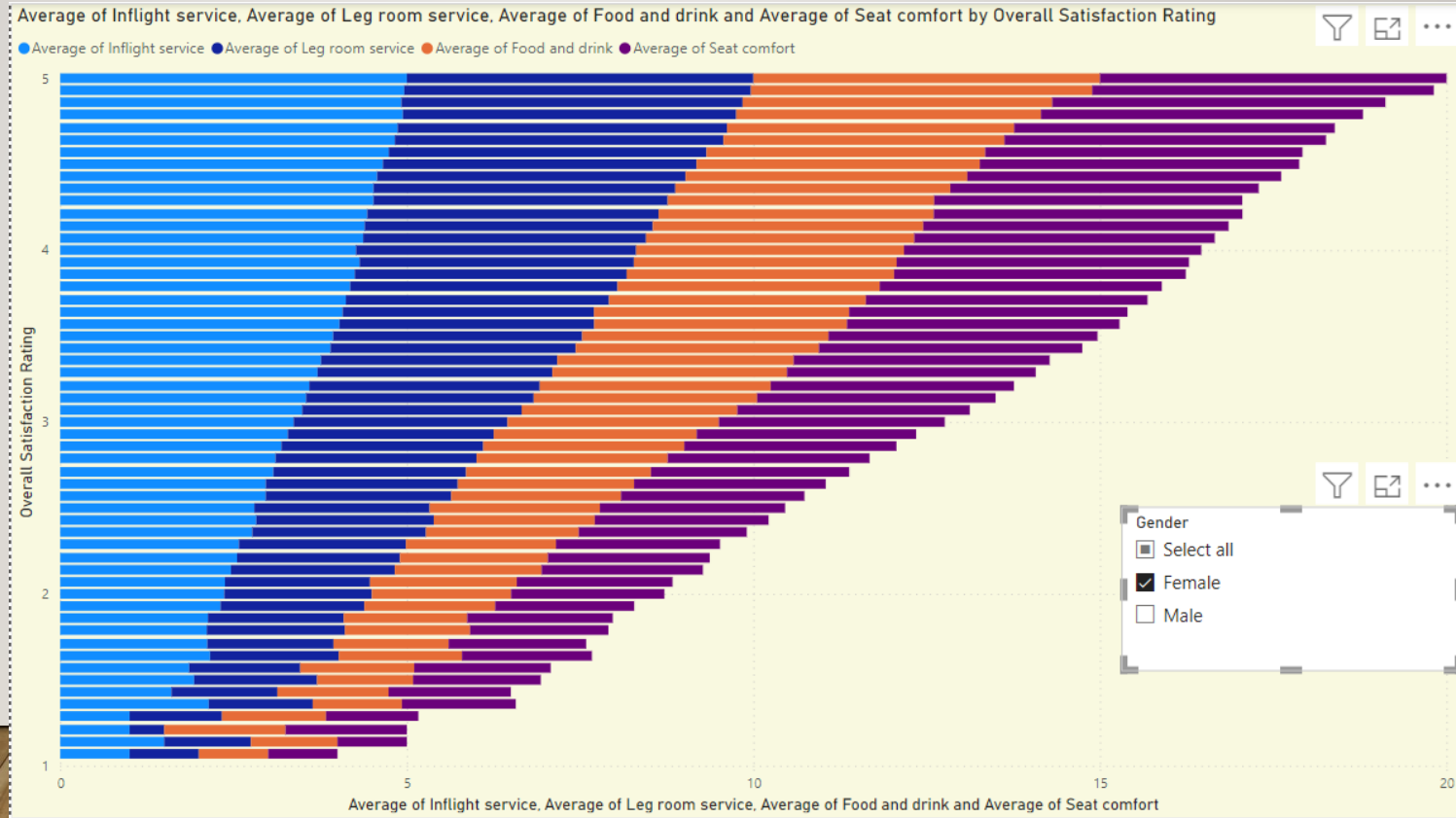
POWERBI

Bar Chart of Average of Certain Factors That Influence Passenger Satisfaction by Gender



POWERBI

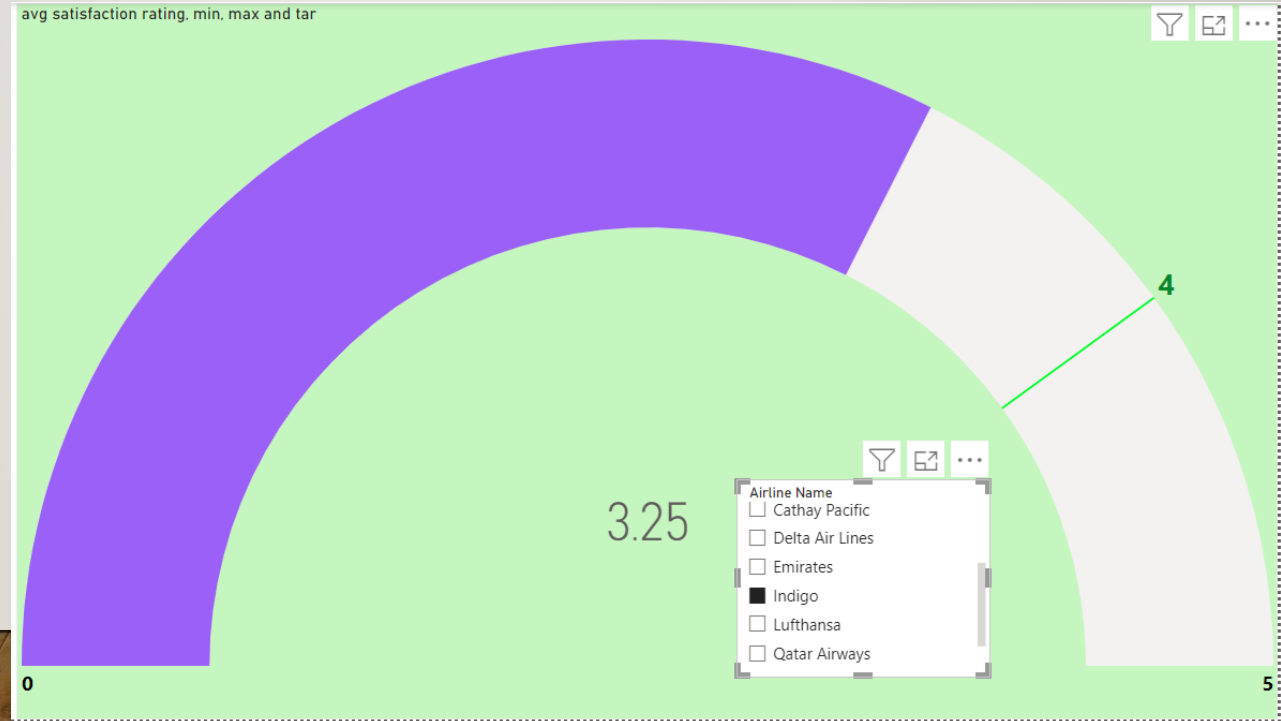
Filter Applied
on Female
selection.



POWERBI

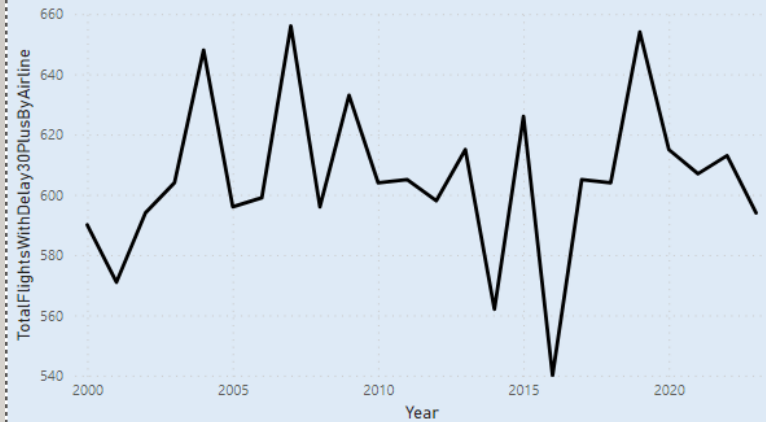
Gauge chart for overall satisfaction rating with airline filter.

The target value is 4

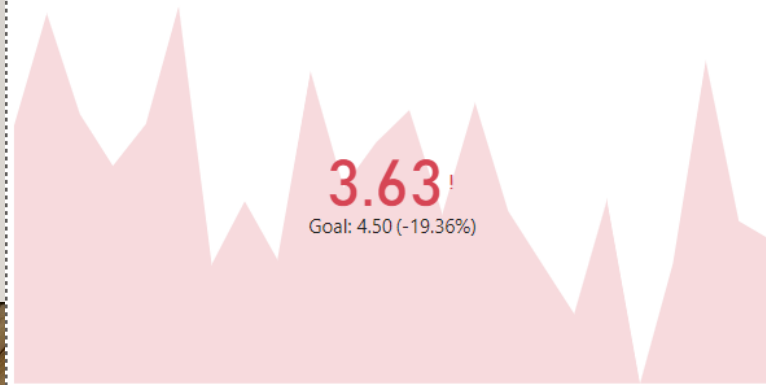


POWERBI DASHBOARD

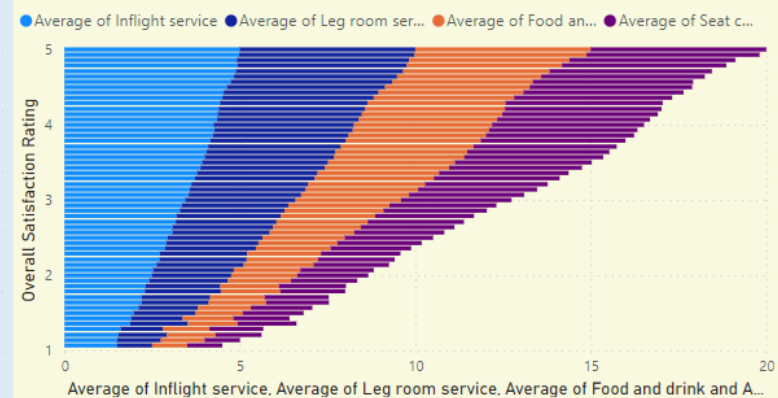
TotalFlightsWithDelay30PlusByAirline by Year



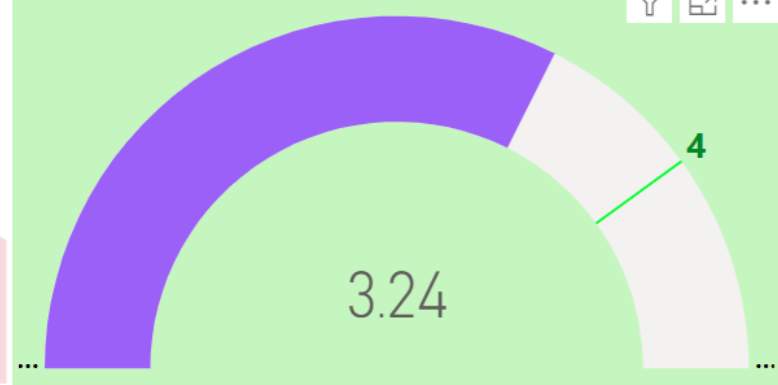
Average of Inflight service and Target KPI by Year



Average of Inflight service, Average of Leg room service, Average of Food and drink and Average of Seat comfort by Overall Satisfaction Rating

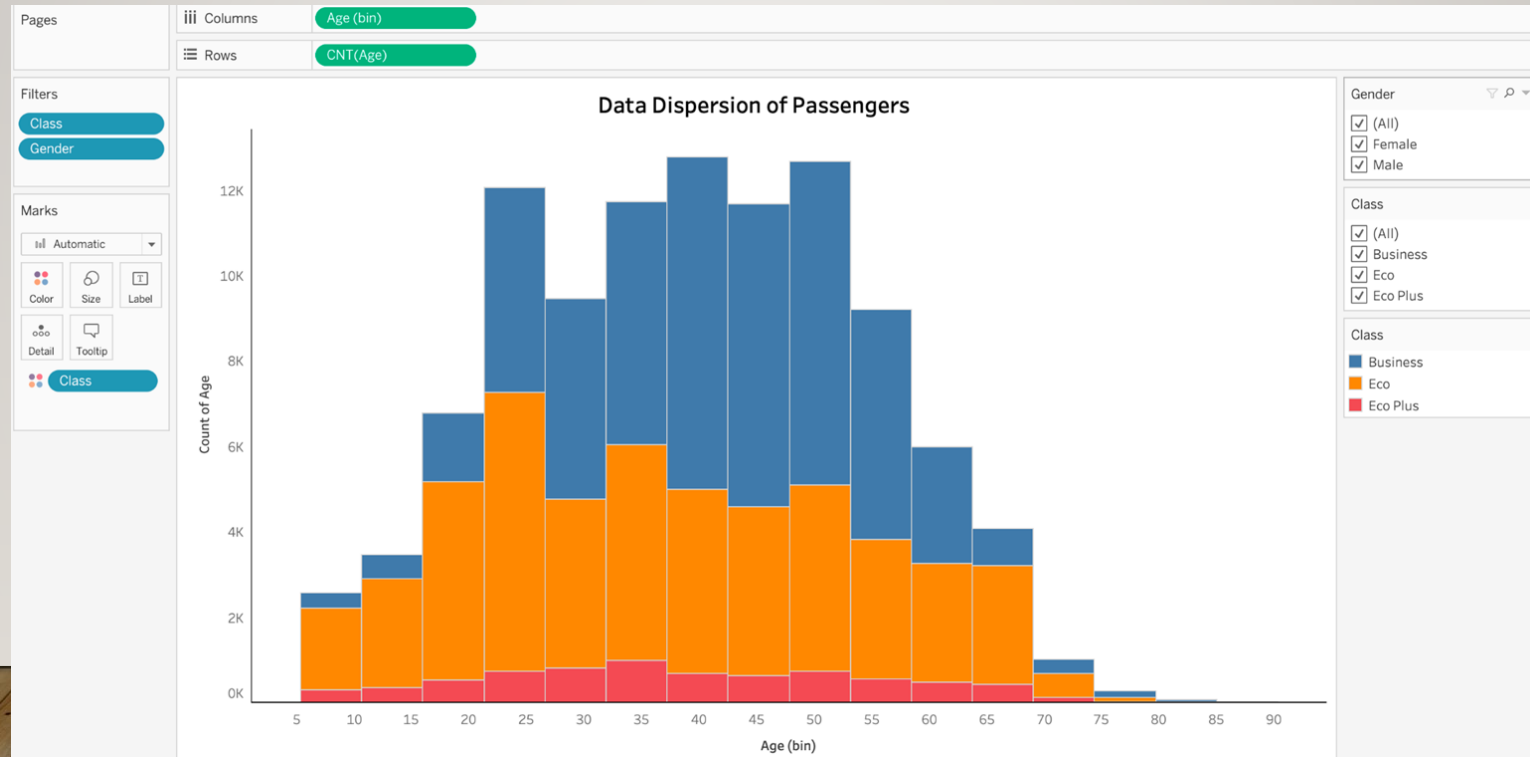


avg satisfaction rating, min, max and tar

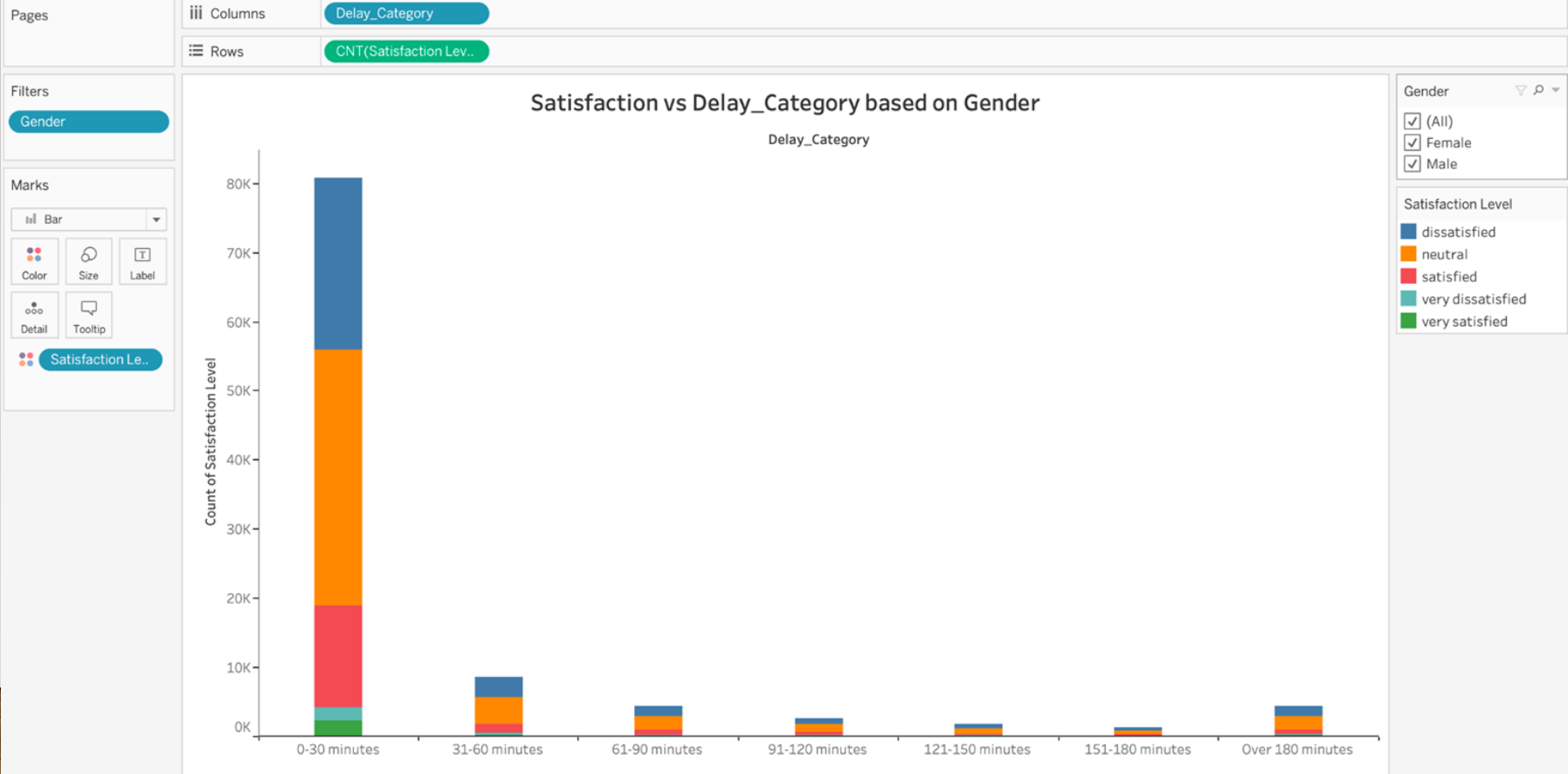


TABLEAU

Data Dispersion of Passengers



Satisfaction vs Delay graph based on gender



TABLEAU

Created two calculated fields called Total_delay and Delay_category.

`[Departure Delay in Minutes] + [Arrival Delay in Minutes]`

The calculation is valid.

3 Dependencies ▾

Apply

OK

```
IF [total_delay] <= 30 THEN '0-30 minutes'
ELSEIF [total_delay] <= 60 THEN '31-60 minutes'
ELSEIF [total_delay] <= 90 THEN '61-90 minutes'
ELSEIF [total_delay] <= 120 THEN '91-120 minutes'
ELSEIF [total_delay] <= 150 THEN '121-150 minutes'
ELSEIF [total_delay] <= 180 THEN '151-180 minutes'
ELSE 'Over 180 minutes'
END
```

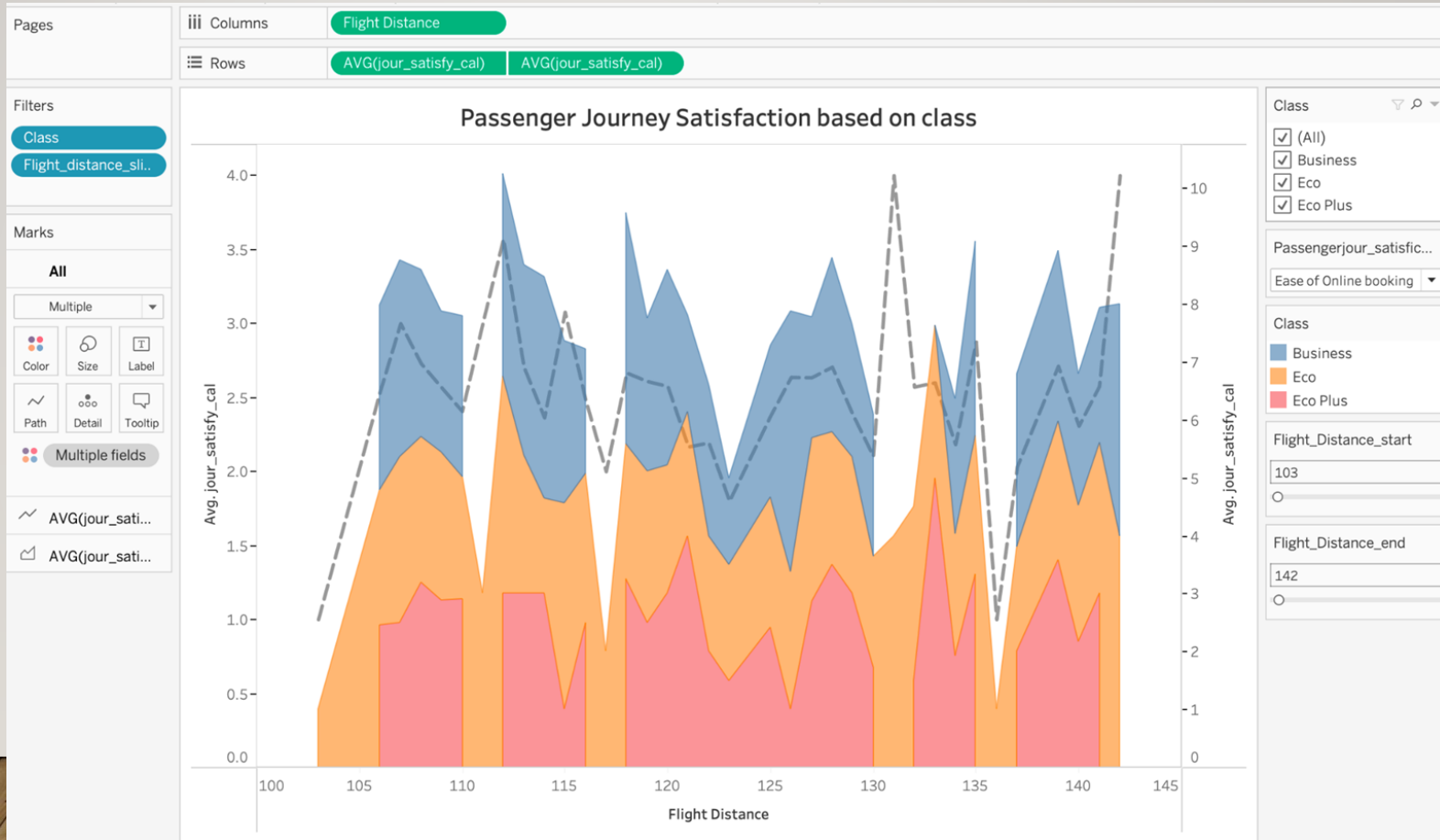
The calculation is valid.

2 Dependencies ▾

Apply

OK

Passenger Journey Satisfaction graph



TABLEAU

For this graph, created a jour_satisfy_cal (parameter), Passengerjour_satisfaction (calculated field) and flight_distance slider.

jour_satisfy_cal

```
CASE [Passengerjour_satisfaction]
WHEN "Cleanliness" THEN [Cleanliness]
WHEN "Leg room service" THEN [Leg room service]
WHEN "Seat comfort" THEN [Seat comfort]
WHEN "Ease of Online booking" THEN [Ease of Online booking]
WHEN "Food and Drink" THEN [Food and drink]

END|
```

The calculation is valid.

2 Dependencies ▼

Apply

OK

Edit Parameter [Passengerjour_satisfaction]

Name

Passengerjour_satisfaction

Properties

Data type

String ▼

Display format

Ease of Online booking ▼

Current value

Ease of Online booking ▼

Value when workbook opens

Current value ▼

Allowable values

☐ All

☒ List

☐ Range

Value	Display As
Cleanliness	Cleanliness
Seat comfort	Seat comfort
Leg room service	Leg room service
Ease of Online booking	Ease of Online booking
Food and Drink	Food and Drink
Click to add	

☒ Fixed

☐ When workbook opens

Add values from ▼

Remove Selected

Cancel

OK

TABLEAU

Flight_distance Sliders

✕

Edit Parameter [Flight_Distance_start]

Name

Properties

Data type
Integer ▼

Display format
103 ▼

Current value
103

Value when workbook opens
Current value ▼

Allowable values
☐ All ☐ List ☒ Range

Range of values
☒ Minimum ☒ Fixed
☒ Maximum ☐ When workbook opens
☐ Step size

✕

Edit Parameter [Flight_Distance_end]

Name

Properties

Data type
Integer ▼

Display format
142 ▼

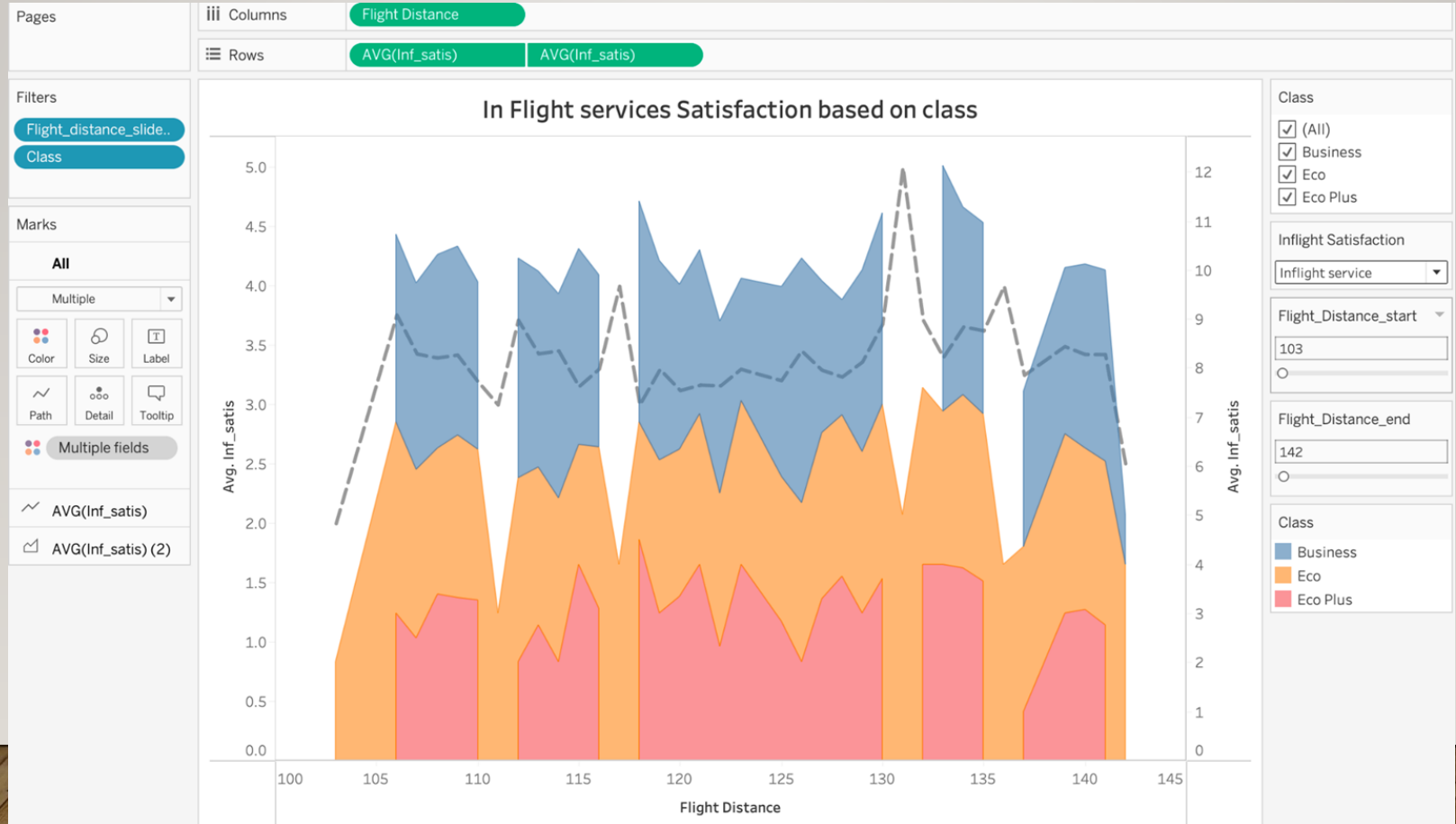
Current value
142

Value when workbook opens
Current value ▼

Allowable values
☐ All ☐ List ☒ Range

Range of values
☒ Minimum ☒ Fixed
☒ Maximum ☐ When workbook opens
☐ Step size

In Flight Service Satisfaction graph



TABLEAU

For this graph, created a
Inf_satis(parameter), and Inflight
Satisfaction (calculated field).

Inf_satis

```
CASE [Inflight Satisfaction]
WHEN "Inflight wifi service" THEN [Inflight wifi service]
WHEN "Inflight entertainment" THEN [Inflight entertainment]
WHEN "Inflight service" THEN [Inflight service]
END
```

The calculation is valid.

2 Dependencies ▼

Apply

OK

Edit Parameter [Inflight Satisfaction]

Name

Inflight Satisfaction

Properties

Data type

String ▼

Display format

Inflight service ▼

Current value

Inflight service ▼

Value when workbook opens

Current value ▼

Allowable values

☐ All ☒ List ☐ Range

Value	Display As
Inflight entertainment	Inflight entertainment
Inflight service	Inflight service
Inflight wifi service	Inflight wifi service
Click to add	

☒ Fixed ☐ When workbook opens

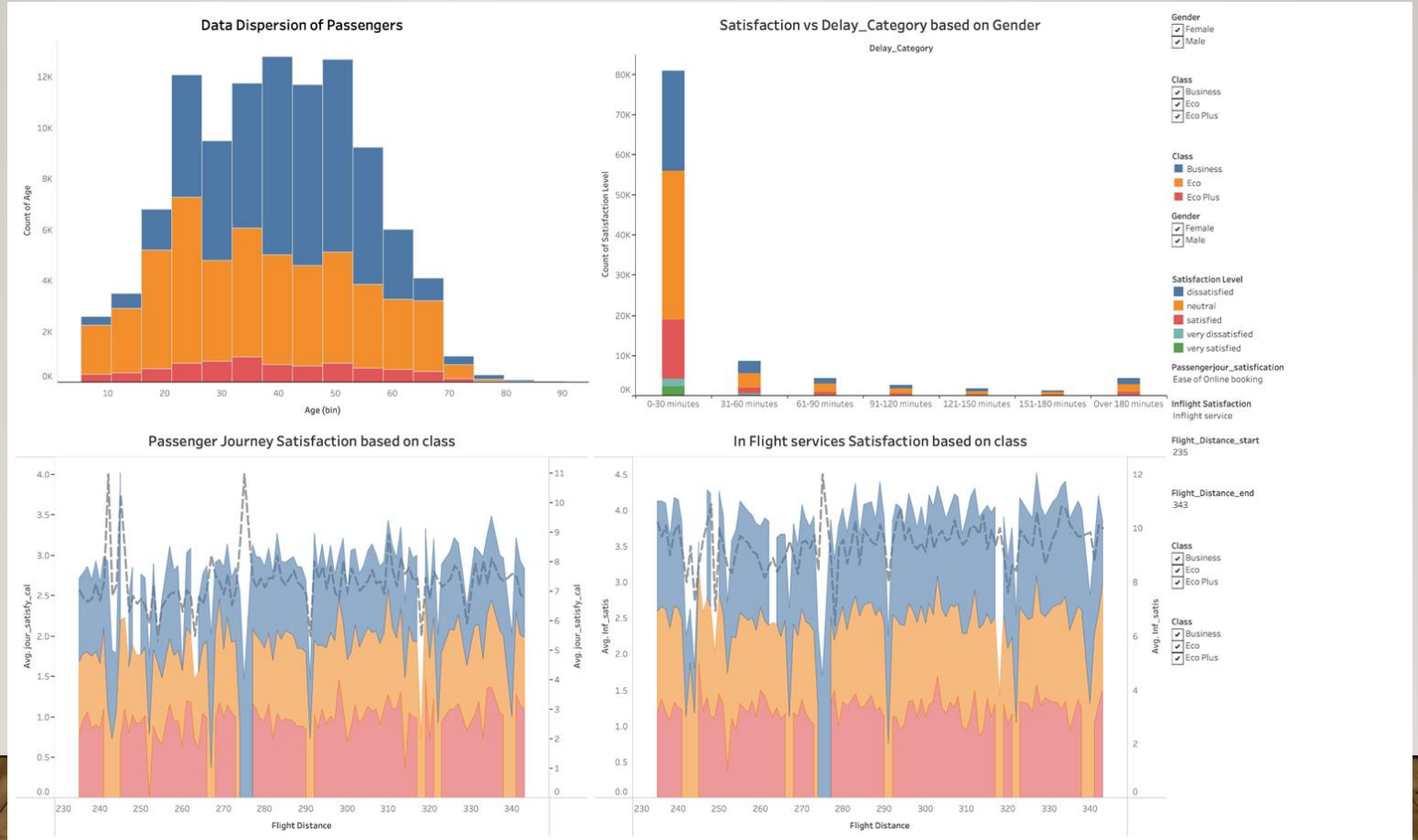
Add values from ▼

Remove Selected

Cancel

OK

TABLEAU DASHBOARD



CONCLUSION

The study conducted a thorough analysis of airline passenger satisfaction, employing tools such as Shiny R, Tableau, and Power BI for a deep dive into the data.

Through interactive visualizations, the study was able to uncover critical trends and correlations that affect passenger experiences in the aviation industry



REFERENCES

<https://www.kaggle.com/datasets/teejmahal20/airline-passenger-satisfaction>

RStudio Team. (2022). Shiny: Web Application Framework for R. R package version 1.7.1. Retrieved from <https://shiny.rstudio.com/>

Tableau Software. (2022). Tableau. Retrieved from <https://www.tableau.com/>

Microsoft. (2022). Power BI. Retrieved from <https://powerbi.microsoft.com/>

